

西南交通大學
Southwest Jiaotong University

Operating System

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Content

1. Processor Management
2. Process
3. Thread
4. **Scheduling （调度）**
5. Concurrency (并发)
6. Synchronization and Deadlock



When to Schedule?

- Creation of a process
- Termination of a process
- Interruption of I/O
- Interruption of clocks

Processor Scheduling

- assign processes to be executed by the processor in a way that meets system objectives, such as response time, throughput, and processor efficiency
- Broken down into three separate functions:
  - long term scheduling
  - medium term scheduling
  - short term scheduling

Long-term scheduling	The decision to add to the pool of processes to be executed
Medium-term scheduling	The decision to add to the number of processes that are partially or fully in main memory
Short-term scheduling	The decision as to which available process will be executed by the processor
I/O scheduling	The decision as to which process's pending I/O request shall be handled by an available I/O device



Scheduling and Process State Transitions

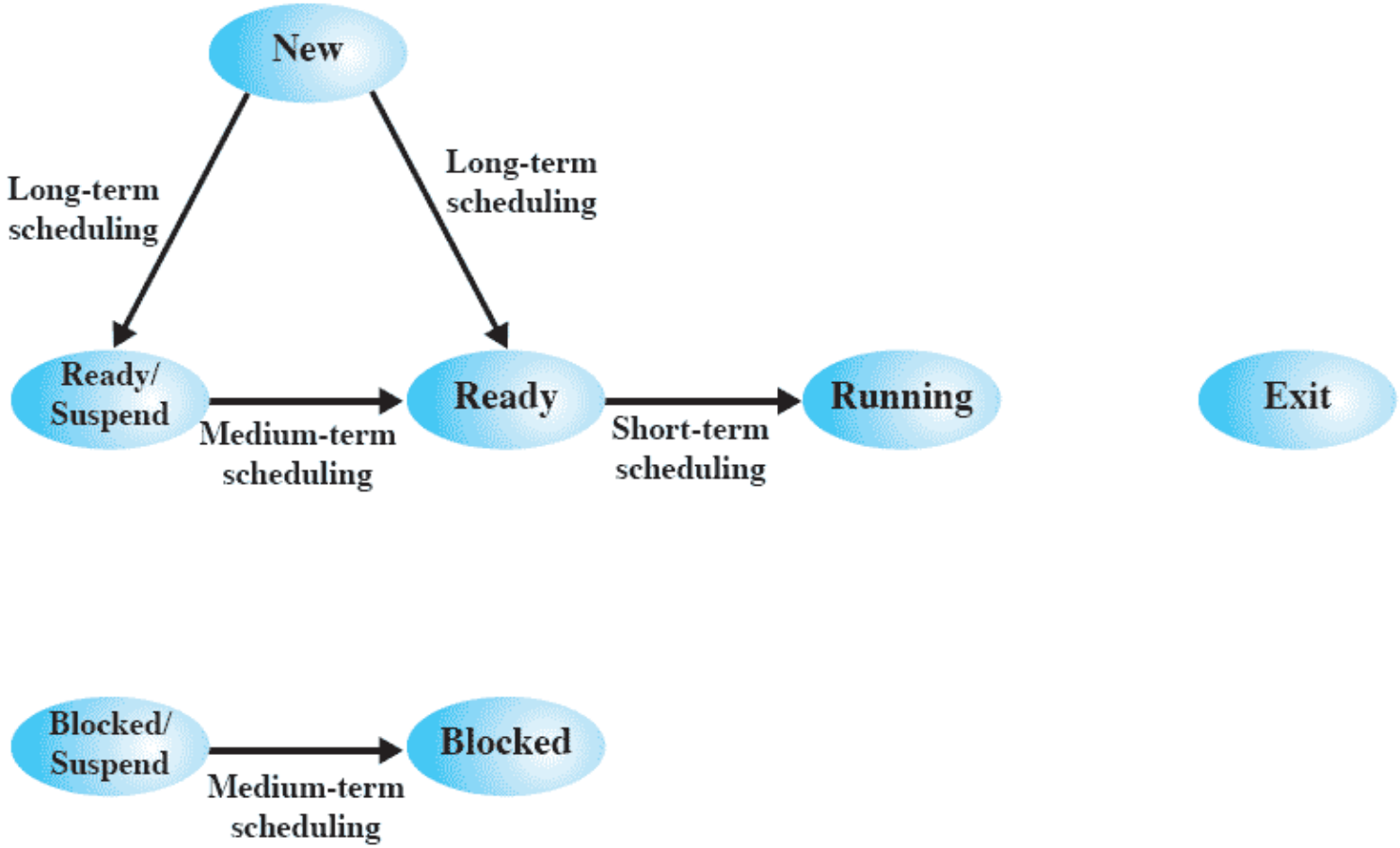
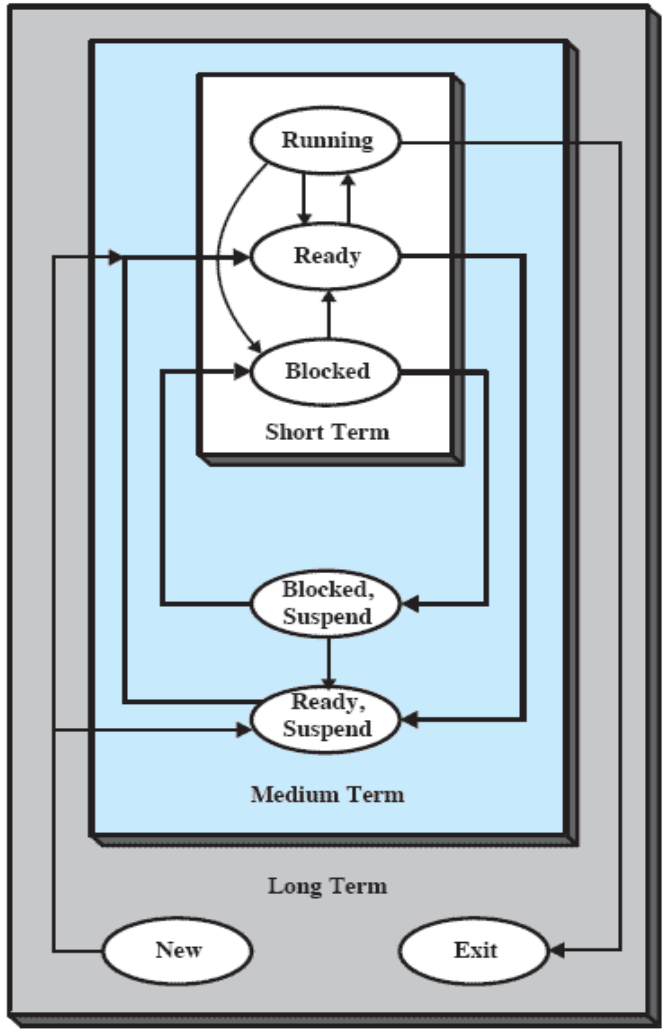


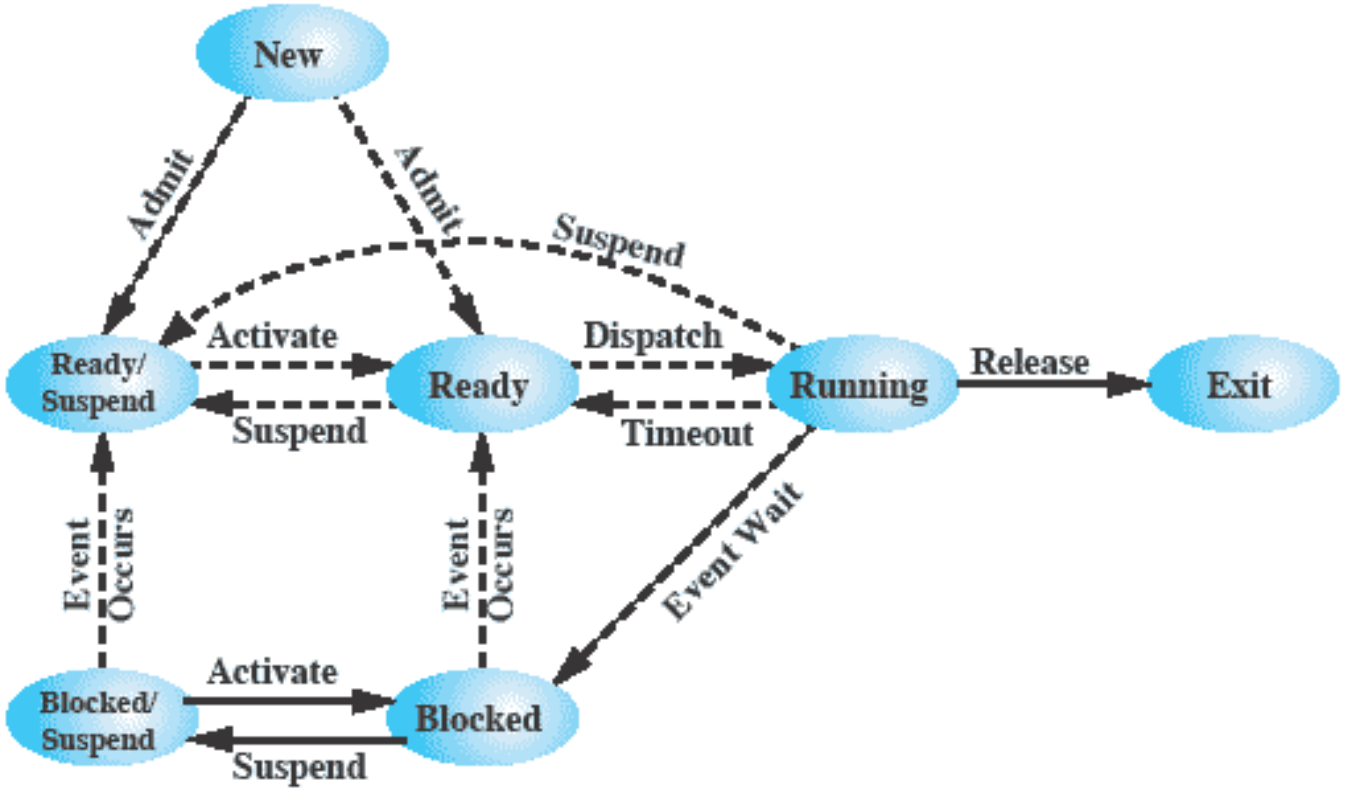
Figure 9.1 Scheduling and Process State Transitions

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Nesting （嵌套） of Scheduling Functions



(Referencing figure 3.9b)



(b) With Two Suspend States

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Queuing Diagram

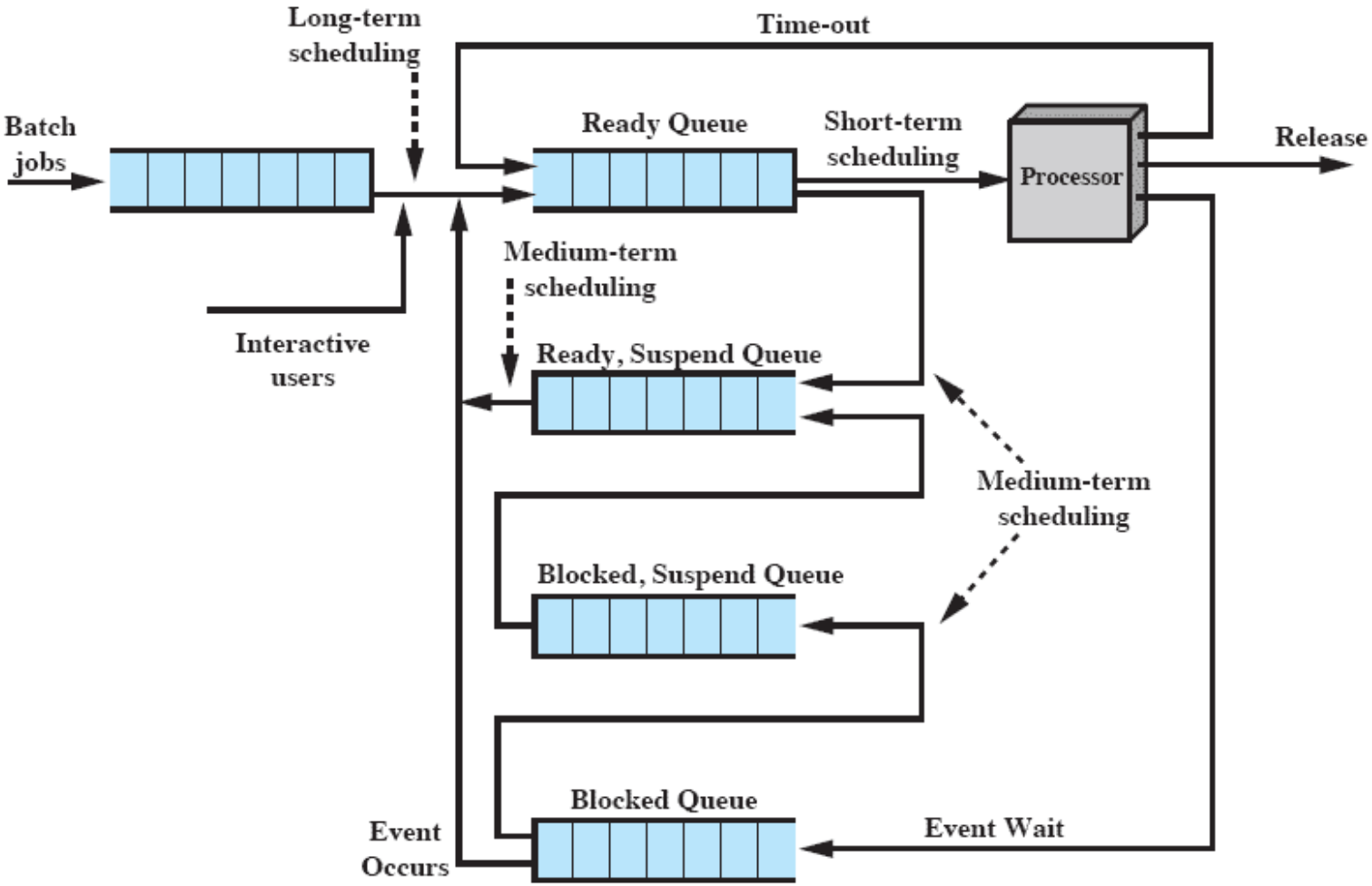


Figure 9.3 Queuing Diagram for Scheduling

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Long-Term Scheduling

- Determines which programs are admitted to the system for processing
- Controls the degree of multiprogramming
  - the more processes that are created, the smaller the percentage of time that each process can be executed
  - may limit to provide satisfactory service to the current set of processes
- Creates processes from the queue when it can, but must decide:
  - when the operating system can take on one or more additional processes
  - which jobs to accept and turn into processes
    - ✓ first come, first served
    - ✓ priority, expected execution time, I/O requirements

Medium-Term Scheduling

- Also called Intermediate-Level Scheduling

- Part of the swapping function
- Swapping-in decisions are based on the need to manage the degree of multiprogramming
  - considers the memory requirements of the swapped-out processes
- To improve memory utilization and system throughput. For this reason, processes that are temporarily unable to run should no longer occupy valuable memory resources, but be transferred to external storage to wait, and the process state at this time should be called the ready external storage state or suspended state.

Short-Term Scheduling

- Known as the dispatcher
- Executes most frequently
- Makes the fine-grained decision of which process to execute next
- Invoked when an event occurs that may lead to the blocking of the current process or that may provide an opportunity to preempt （抢占） a currently running process in favor of another
- Examples:
  - Clock interrupts
  - I/O interrupts
  - Operating system calls
  - Signals (e.g., semaphores)

Short Term Scheduling Criteria

- User-oriented
  - relate to the behavior of the system as perceived （感知） by the individual user or process (such as response time in an interactive system)
  - important on virtually all systems
- System-oriented
  - focus in on effective and efficient utilization of the processor (rate at which processes are completed)
  - generally of minor importance on single-user systems
- Performance-related (examples: response time, throughput)
  - quantitative
  - easily measured
- Non-performance related (examples: predictability)
  - qualitative
  - hard to measure

Scheduling Criteria

User Oriented, Performance Related

Turnaround time This is the interval of time between the submission of a process and its completion. Includes actual execution time plus time spent waiting for resources, including the processor. This is an appropriate measure for a batch job.

Response time For an interactive process, this is the time from the submission of a request until the response begins to be received. Often a process can begin producing some output to the user while continuing to process the request. Thus, this is a better measure than turnaround time from the user's point of view. The scheduling discipline should attempt to achieve low response time and to maximize the number of interactive users receiving acceptable response time.

Deadlines When process completion deadlines can be specified, the scheduling discipline should subordinate other goals to that of maximizing the percentage of deadlines met.

User Oriented, Other

Predictability A given job should run in about the same amount of time and at about the same cost regardless of the load on the system. A wide variation in response time or turnaround time is distracting to users. It may signal a wide swing in system workloads or the need for system tuning to cure instabilities.

System Oriented, Performance Related

Throughput The scheduling policy should attempt to maximize the number of processes completed per unit of time. This is a measure of how much work is being performed. This clearly depends on the average length of a process but is also influenced by the scheduling policy, which may affect utilization.

Processor utilization This is the percentage of time that the processor is busy. For an expensive shared system, this is a significant criterion. In single-user systems and in some other systems, such as real-time systems, this criterion is less important than some of the others.

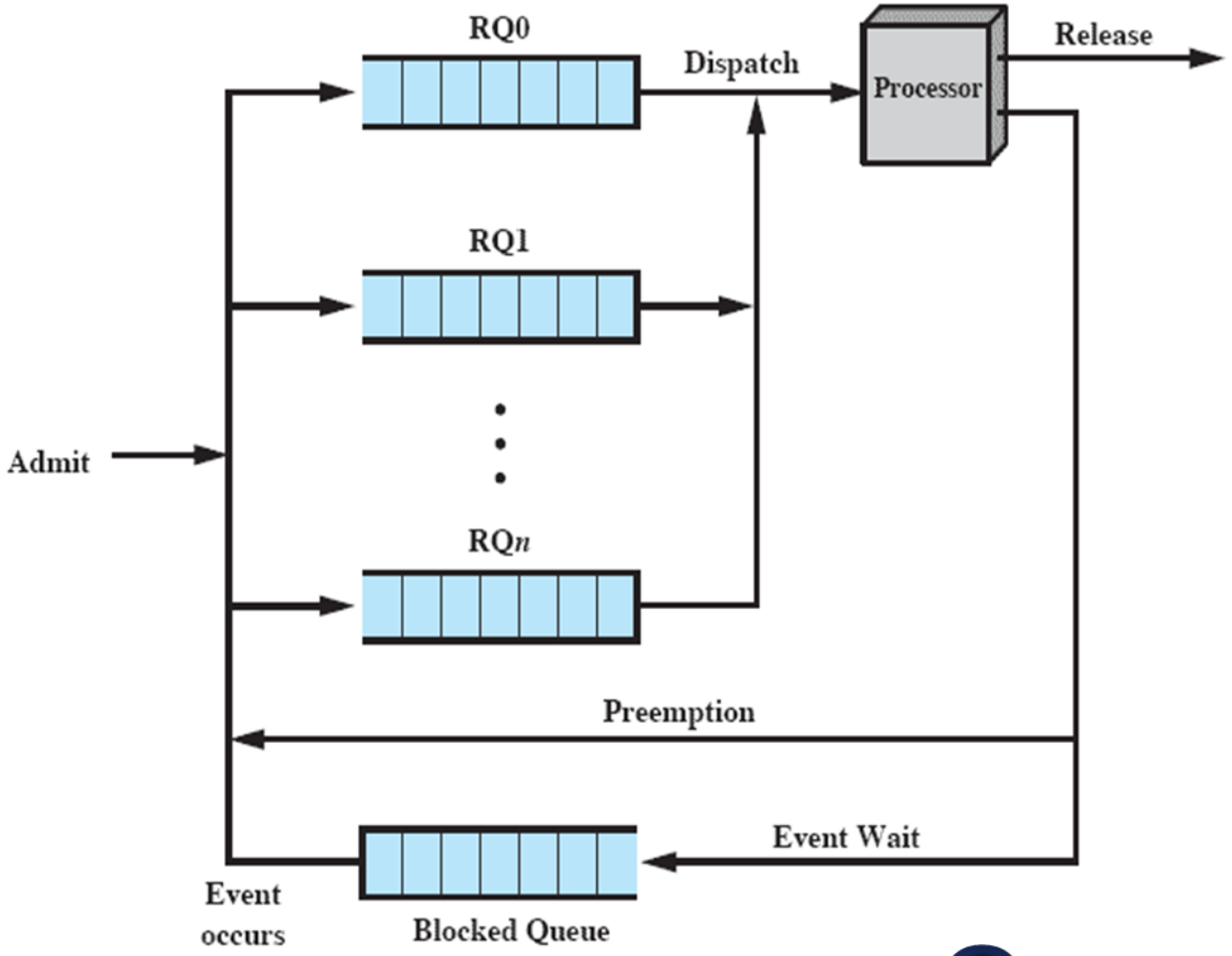
System Oriented, Other

Fairness In the absence of guidance from the user or other system-supplied guidance, processes should be treated the same, and no process should suffer starvation.

Enforcing priorities When processes are assigned priorities, the scheduling policy should favor higher-priority processes.

Balancing resources The scheduling policy should keep the resources of the system busy. Processes that will underutilize stressed resources should be favored. This criterion also involves medium-term and long-term scheduling.

Priority Queuing



? Disadvantage

Figure 9.4 Priority Queuing

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Selection Function (选择函数)

- Determines which process, among ready processes, is selected next for execution
- May be based on priority, resource requirements, or the execution characteristics of the process
- If based on execution characteristics then important quantities are:
  - w = time spent in system so far, waiting
  - e = time spent in execution so far
  - s = total service time required by the process, including e; generally, this quantity must be estimated or supplied by the user

Decision Mode（决策模式）

- Specifies the instants in time at which the selection function is exercised
- Two categories:
  - Nonpreemptive （非抢占式）
  - Preemptive （抢占式）: The purpose is to prevent a process from occupying the CPU



Round Robin?

Principle of Preemptive

- The principle of priority.
- The principle of prioritizing short assignments (processes).
- Time slice principle.



Round Robin?

Nonpreemptive vs Preemptive

- Nonpreemptive
  - once a process is in the running state, it will continue until it terminates or blocks itself for I/O
- Preemptive(抢占)
  - currently running process may be interrupted and moved to ready state by the OS
  - preemption may occur when new process arrives, on an interrupt, or periodically
  - General premise: Predicted running time

Types of Priority

- Static priority: is determined at the time of process creation and remains unchanged throughout the entire runtime of the process (Some systems use "0" to indicate the highest priority)
- Dynamic priority: The priority assigned when creating a process can be changed as the process progresses or as its waiting time increases, in order to achieve better scheduling performance

General basis for determining process priority

- Process type
- The demand for resources by processes
- User requirements

Alternative Scheduling Policies

Table 9.3 Characteristics of Various Scheduling Policies

	FCFS	Round robin	SPN	SRT	HRRN	Feedback
Selection function	max[w]	constant	min[s]	min[s − e]	$\max\left(\frac{w+s}{s}\right)$	(see text)
Decision mode	Non-preemptive	Preemptive (at time quantum)	Non-preemptive	Preemptive (at arrival)	Non-preemptive	Preemptive (at time quantum)
Throughput	Not emphasized	May be low if quantum is too small	High	High	High	Not emphasized
Response time	May be high, especially if there is a large variance in process execution times	Provides good response time for short processes	Provides good response time for short processes	Provides good response time	Provides good response time	Not emphasized
Overhead	Minimum	Minimum	Can be high	Can be high	Can be high	Can be high
Effect on processes	Penalizes short processes; penalizes I/O bound processes	Fair treatment	Penalizes long processes	Penalizes long processes	Good balance	May favor I/O bound processes
Starvation	No	No	Possible	Possible	No	Possible

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Table 9.4 Process Scheduling Example

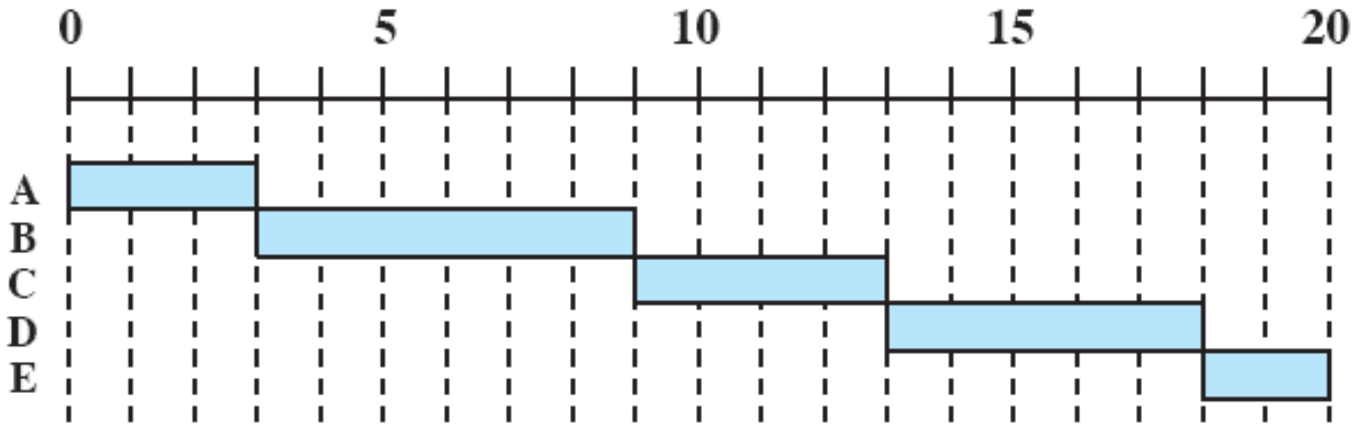
Process	Arrival Time	Service Time
A	0	3
B	2	6
C	4	4
D	6	5
E	8	2

First-Come-First-Served (FCFS)

Process	Arrival Time	Service Time
A	0	3
B	2	6
C	4	4
D	6	5
E	8	2

- Simplest scheduling policy
- Also known as first-in-first-out (FIFO) or a strict queuing scheme
- When the current process ceases to execute, the longest process in the Ready queue is selected
- Performs much better for long processes than short ones
- Tends to favor processor-bound processes over I/O-bound processes

First-Come-First Served (FCFS)



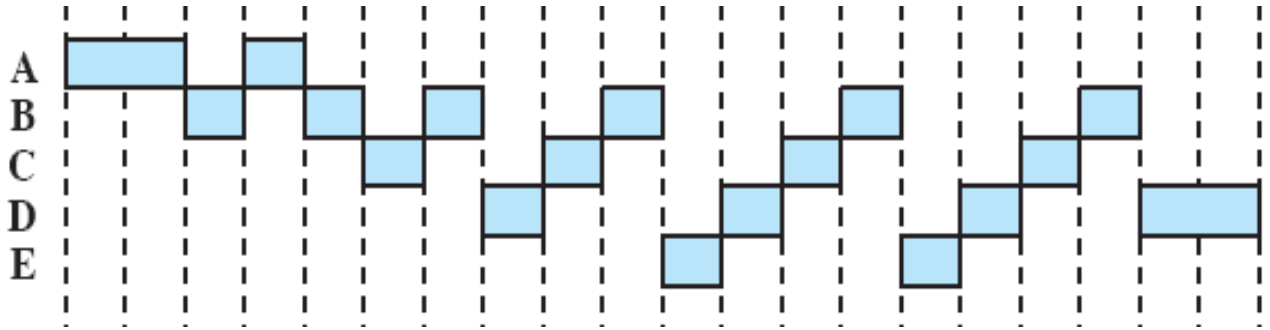
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Round Robin

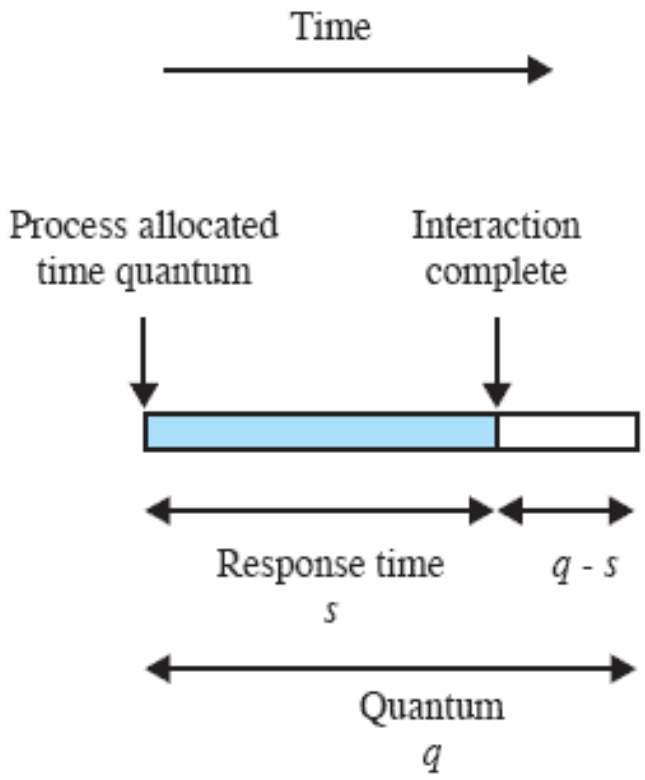
Process	Arrival Time	Service Time
A	0	3
B	2	6
C	4	4
D	6	5
E	8	2

- Uses preemption based on a clock
- Also known as time slicing because each process is given a slice of time before being preempted
- Principal design issue is the length of the time quantum, or slice, to be used
- Particularly effective in a general-purpose time-sharing system or transaction processing system
- One drawback is its relative treatment of processor-bound and I/O-bound processes

Round-Robin (RR), $q = 1$



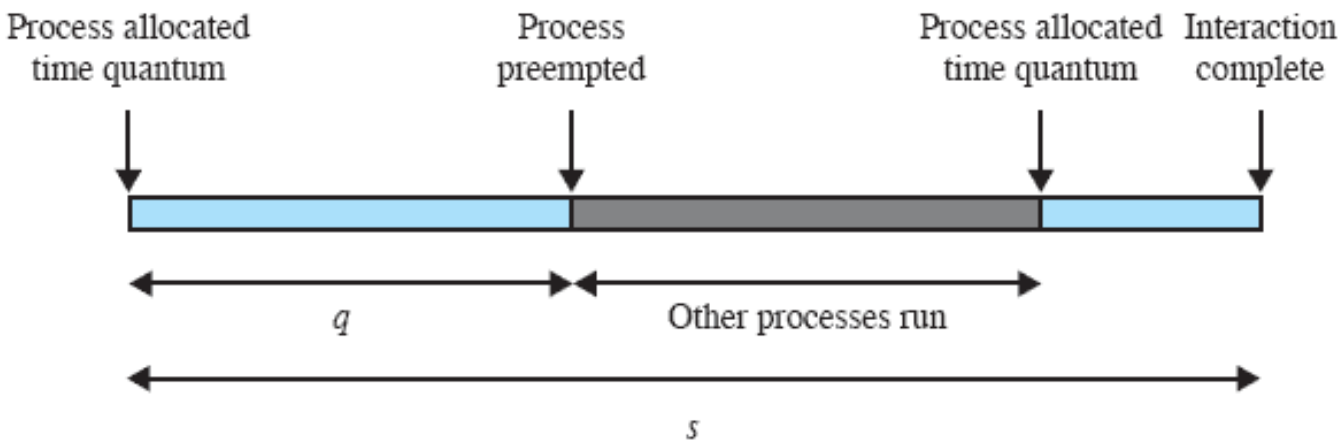
Effect of Size of Preemption Time Quantum



(a) Time quantum greater than typical interaction

Figure 9.6a

Effect of Size of Preemption Time Quantum



(b) Time quantum less than typical interaction

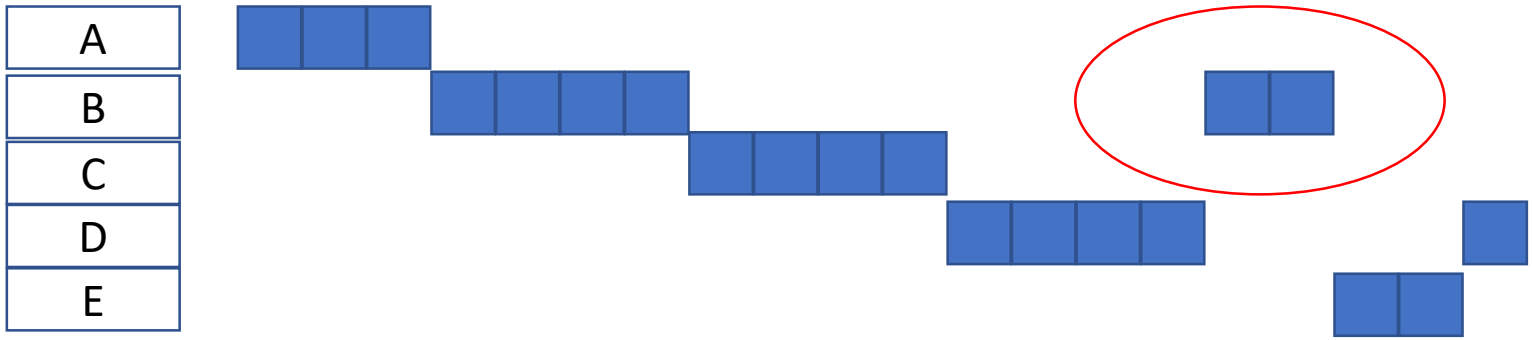
Figure 9.6 Effect of Size of Preemption Time Quantum

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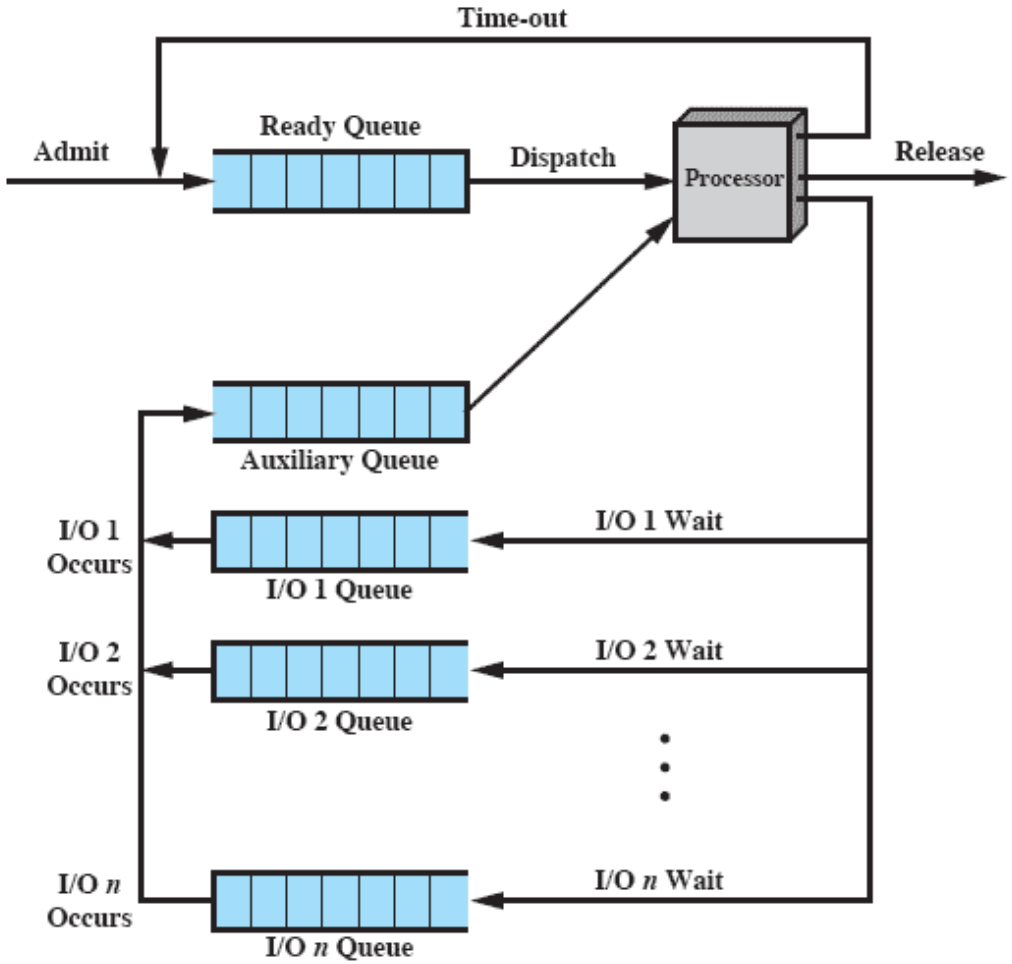
Round Robin

- q=4?

Process	Arrival Time	Service Time
A	0	3
B	2	6
C	4	4
D	6	5
E	8	2



Virtual Round Robin (VRR)



- When a process is blocked for I/O, it joins an I/O queue. Processes that have cleared I/O blocking are transferred to an FCFS auxiliary queue

Figure 9.7 Queuing Diagram for Virtual Round-Robin Scheduler

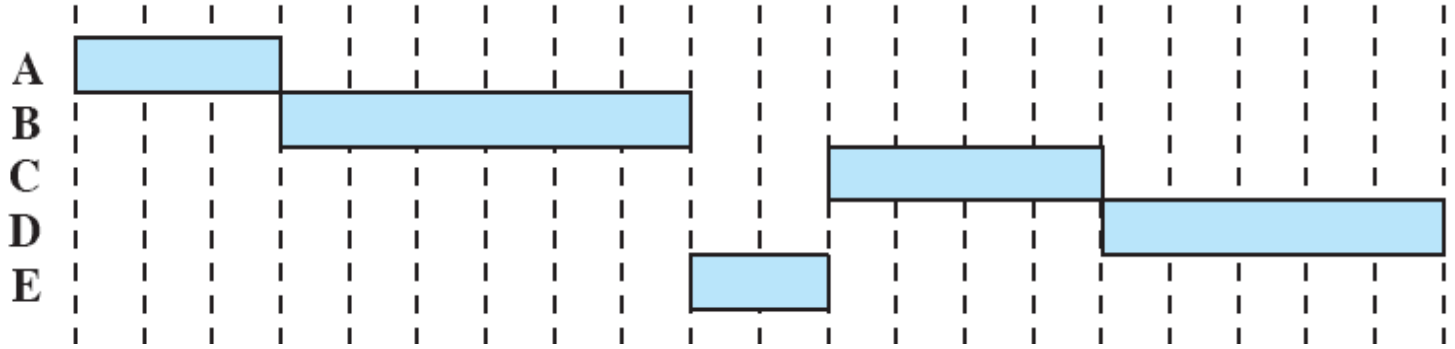
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Shortest Process Next (SPN)

Process	Arrival Time	Service Time
A	0	3
B	2	6
C	4	4
D	6	5
E	8	2

- Also called shortest job first (SJF) or shortest process first (SPF)
- Nonpreemptive policy in which the process with the shortest expected processing time is selected next
- A short process will jump to the head of the queue
- Possibility of starvation for longer processes
- One difficulty is the need to know, or at least estimate, the required processing time of each process
- If the programmer's estimate is substantially under the actual running time, the

Shortest Process Next (SPN)



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Turnaround Time (TAT)

- TAT = waiting time ＋ service time
- Normalized TAT = TAT/service time



Estimate running time

- Simple averaging

$$S_{n+1} = \frac{1}{n}\sum T_i$$

- Exponential averaging

$$S_{n+1} = \alpha T_n + (1-\alpha) S_n$$

$$S_{n+1} = \alpha T_n + (1-\alpha)\alpha T_{n-1} .... + (1-\alpha)^n S_1$$

If n processes have been executed and their actual burst time is Ti, predict the burst time of the next process. Sn is the actual burst time

Exponential Smoothing Coefficients

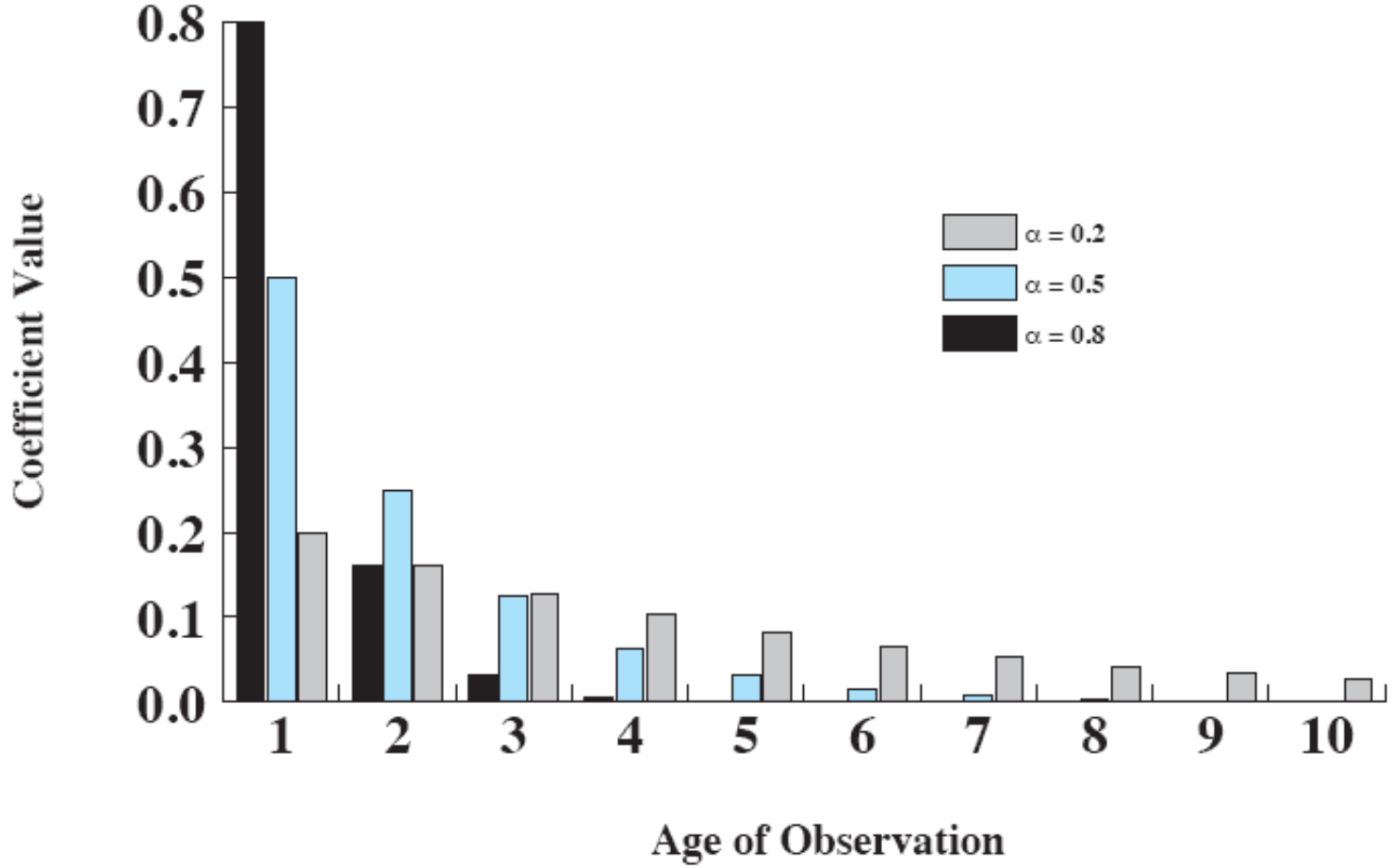
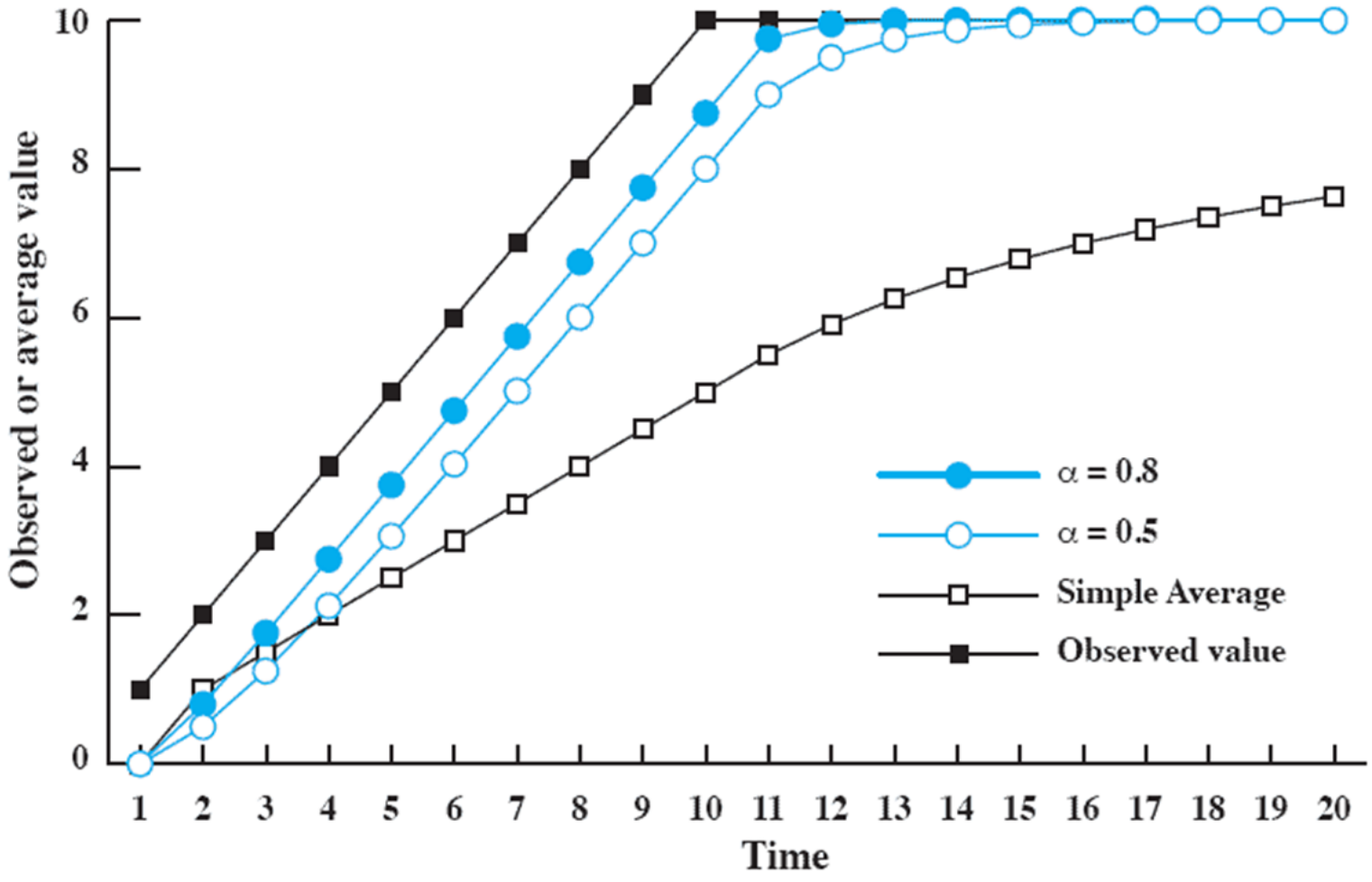


Figure 9.8 Exponential Smoothing Coefficients

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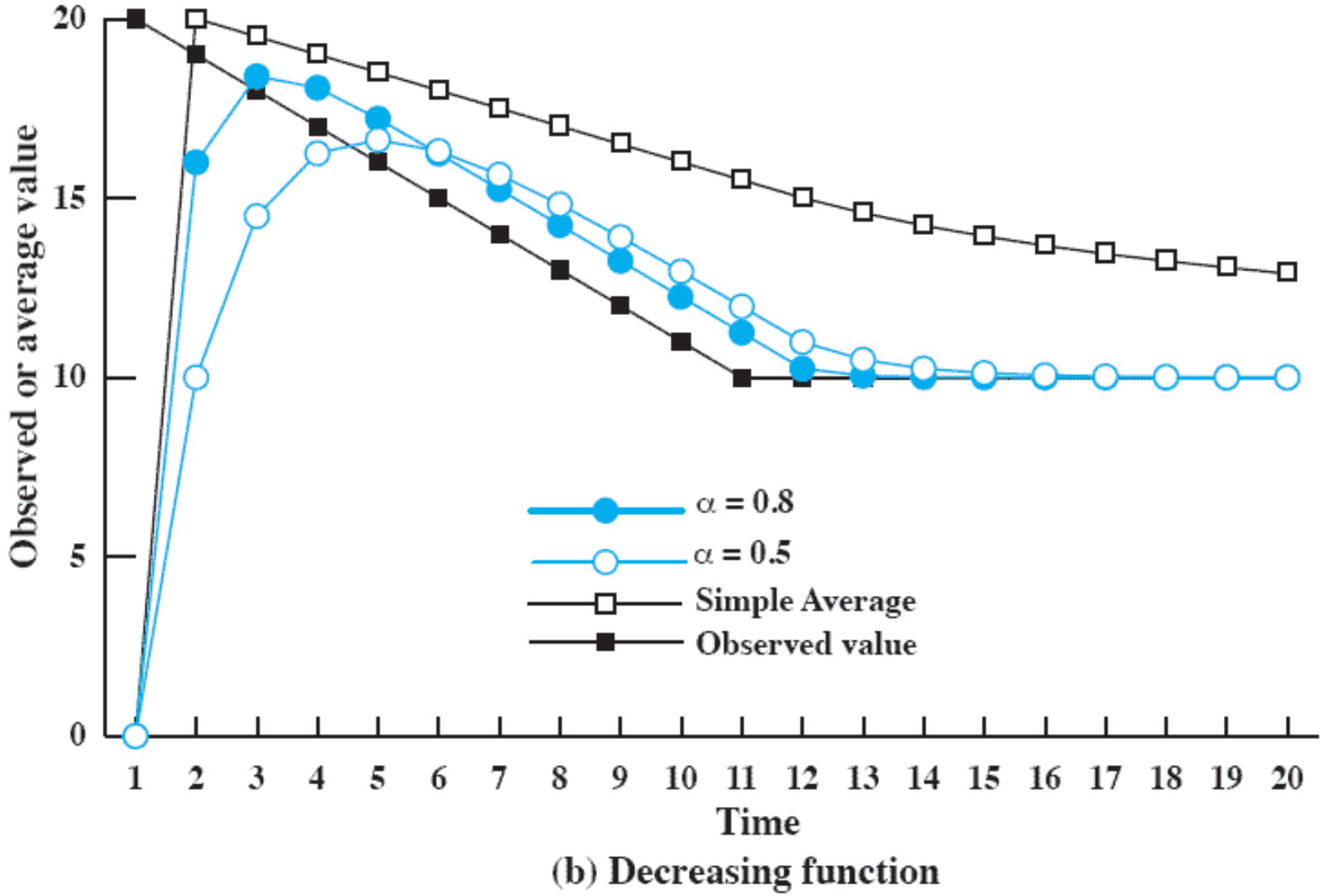
Use Of Exponential Averaging



(a) Increasing function

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Use Of Exponential Averaging

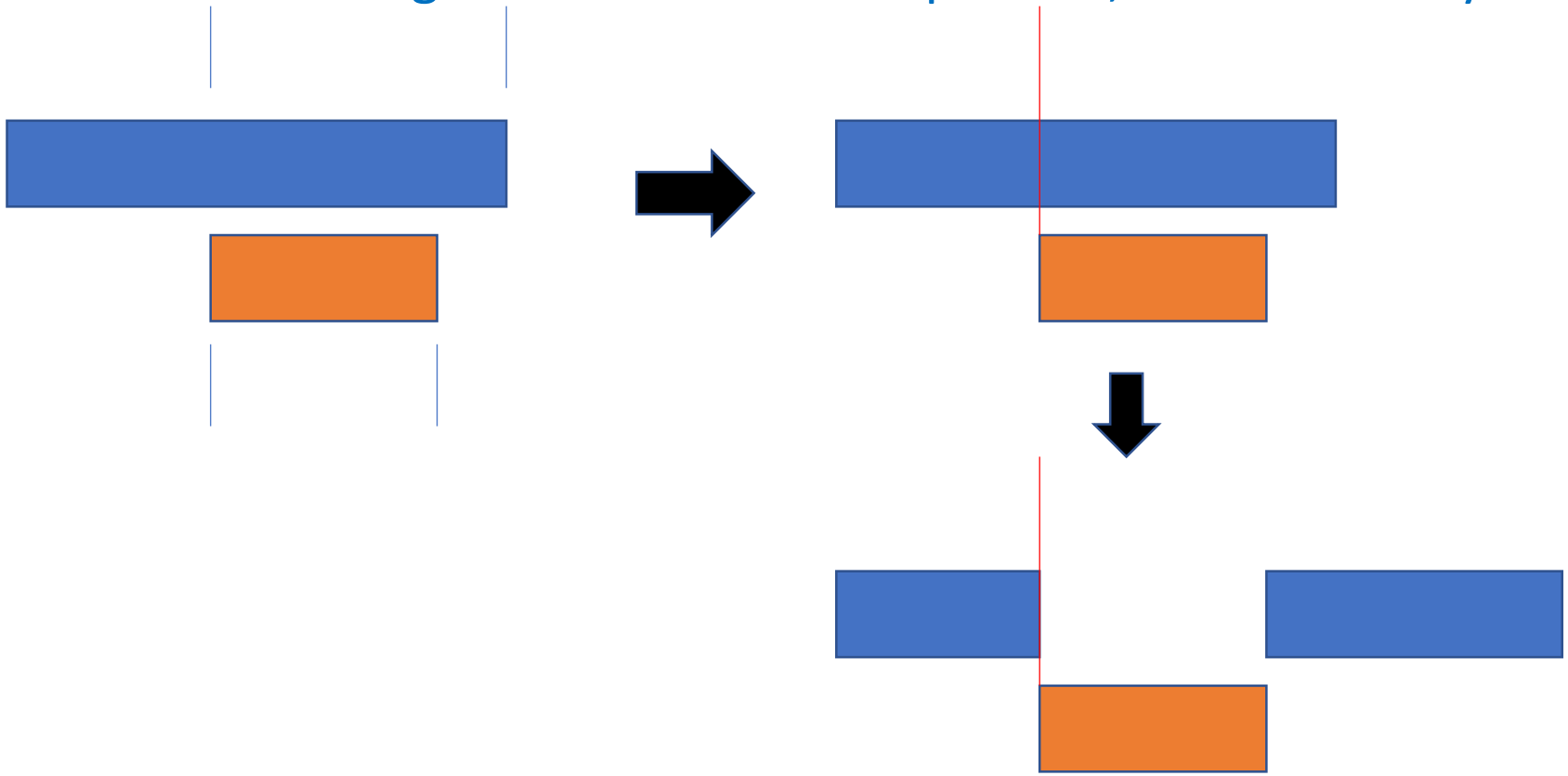


(b) Decreasing function

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Shortest Remaining Time (SRT) 最短剩余时间优先

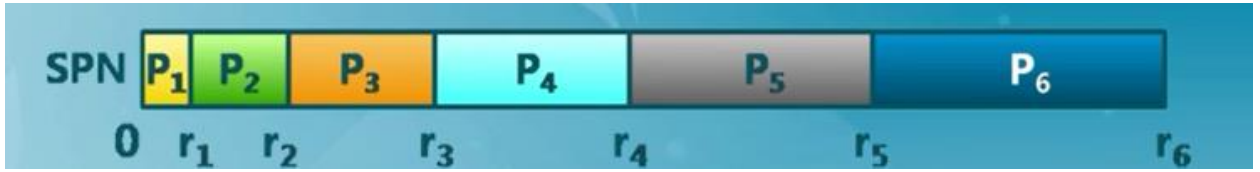
- Its core mechanism is: when a new process arrives, if its remaining time is less than the remaining time of the current process, it immediately seizes CPU control



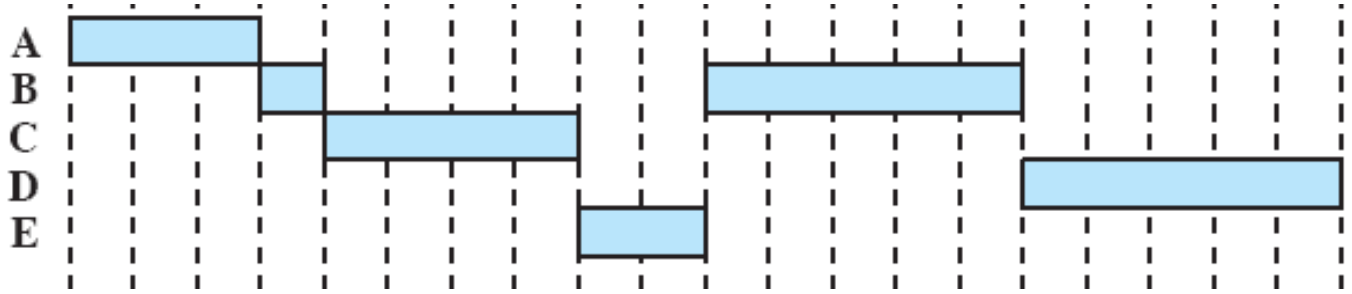
Shortest Remaining Time (SRT) 最短剩余时间优先

Process	Arrival Time	Service Time
A	0	3
B	2	6
C	4	4
D	6	5
E	8	2

- Also called SRTF or SRTN
- Preemptive version of SPN
- Scheduler always chooses the process that has the shortest expected remaining processing time
- Risk of starvation of longer processes
- Should give superior turnaround time performance to SPN because a short job is given immediate preference to a running longer job



Shortest Remaining Time (SRT)



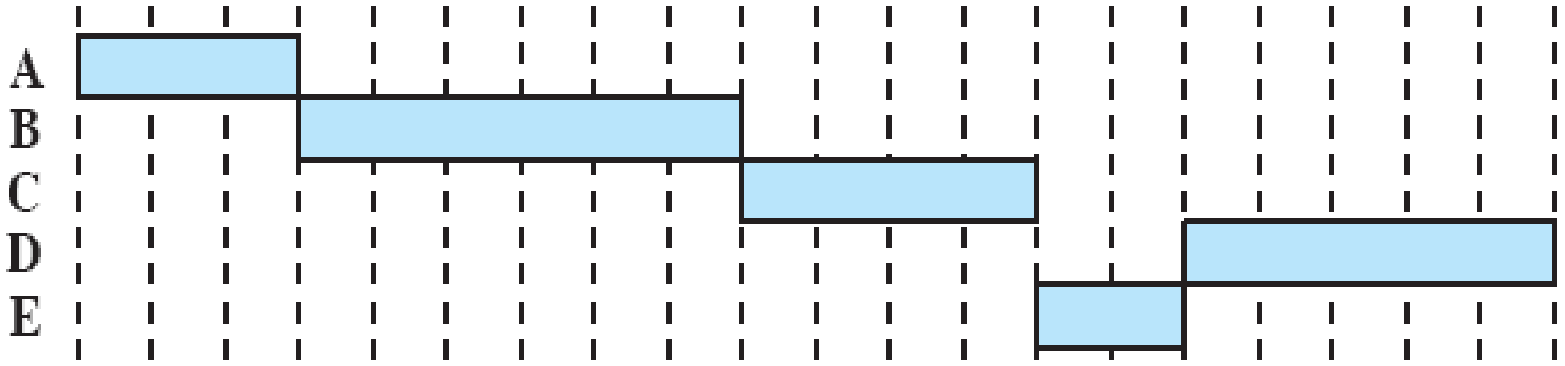
Highest Response Ratio Next (HRRN)最高响应比

Process	Arrival Time	Service Time
A	0	3
B	2	6
C	4	4
D	6	5
E	8	2

- Chooses next process with the greatest ratio
- Attractive because it accounts for the age of the process
- While shorter jobs are favored, aging without service increases the ratio so that a longer process will eventually get past competing shorter jobs

$$Ratio = \frac{time\ spent\ waiting + expected\ service\ time}{expected\ service\ time}$$

Highest Response Ratio Next (HRRN)



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Feedback Scheduling

Multi level feedback queue scheduling algorithm

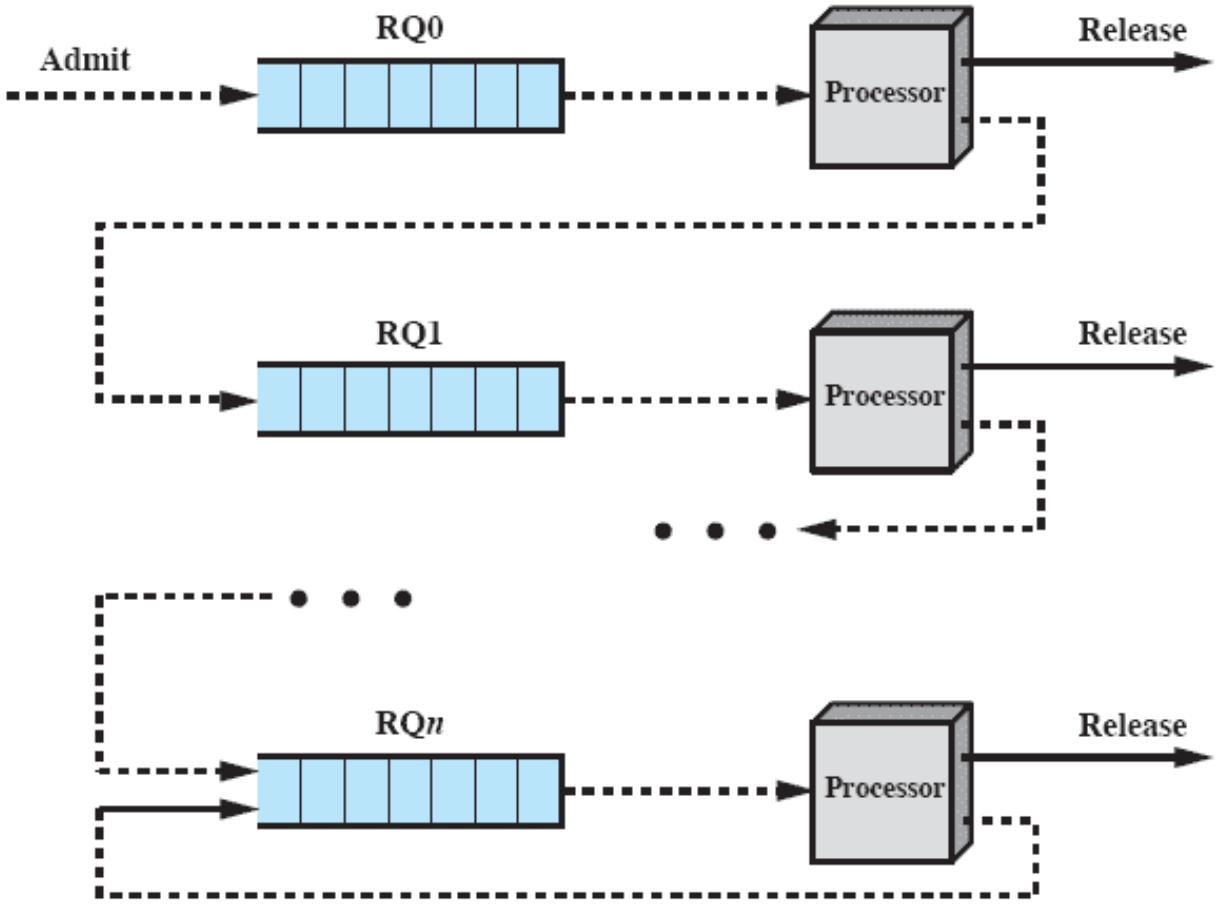


Figure 9.10 Feedback Scheduling

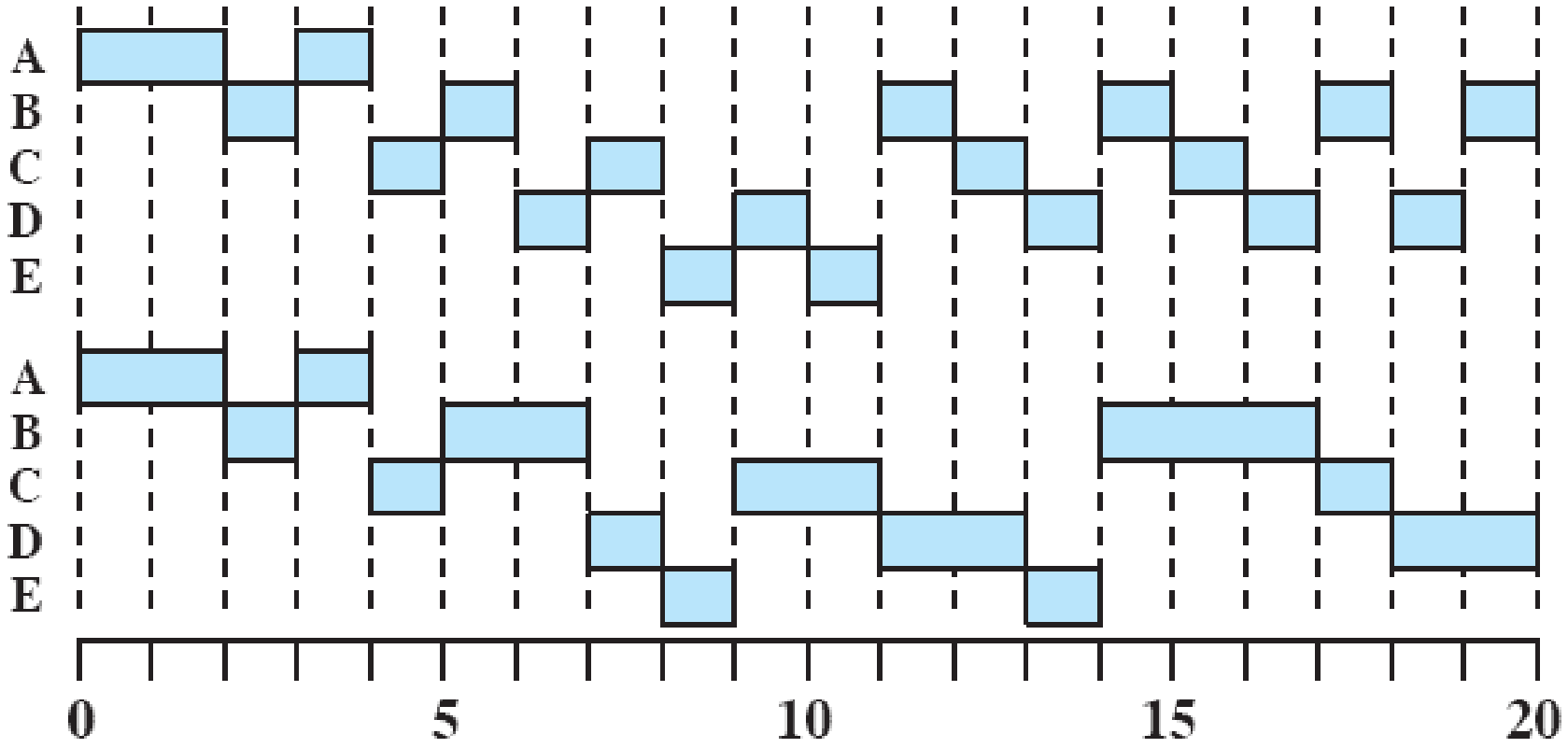
Balancing the need for high priority task response and quick completion of short assignments

S1<S2<…<Sn

Feedback Performance

Feedback
$q = 1$

Feedback
$q = 2^i$



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Performance Comparison

- Any scheduling discipline that chooses the next item to be served independent of service time obeys the relationship:

$$\frac{T_r}{T_s} = \frac{1}{1 - \rho}$$

where

T_r = turnaround time or residence time; total time in system, waiting plus execution

T_s = average service time; average time spent in Running state

ρ = processor utilization

Formulas for Single-Server Queues with Two Priority

Assumptions:
1. Poisson arrival rate.
2. Priority 1 items are serviced before priority 2 items.
3. First-come-first-served dispatching for items of equal priority.
4. No item is interrupted while being served.
5. No items leave the queue (lost calls delayed).

(a) General formulas

$$\lambda = \lambda_1 + \lambda_2$$

$$\rho_1 = \lambda_1 T_{s1};\ \rho_2 = \lambda_2 T_{s2}$$

$$\rho = \rho_1 + \rho_2$$

$$T_s = \frac{\lambda_1}{\lambda} T_{s1} + \frac{\lambda_2}{\lambda} T_{s2}$$

$$T_r = \frac{\lambda_1}{\lambda} T_{r1} + \frac{\lambda_2}{\lambda} T_{r2}$$

(b) No interrupts; exponential service times

$$T_{r1} = T_{s1} + \frac{\rho_1 T_{s1} + \rho_2 T_{s2}}{1 - \rho_1}$$

$$T_{r2} = T_{s2} + \frac{T_{r1} - T_{s1}}{1 - \rho}$$

(c) Preemptive-resume queueing discipline; exponential service times

$$T_{r1} = T_{s1} + \frac{\rho_1 T_{s1}}{1 - \rho_1}$$

$$T_{r2} = T_{s2} + \frac{1}{1 - \rho_1}\left(\rho_1 T_{s2} + \frac{\rho T_s}{1 - \rho}\right)$$

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Normalized Response Time for Shorter Processes

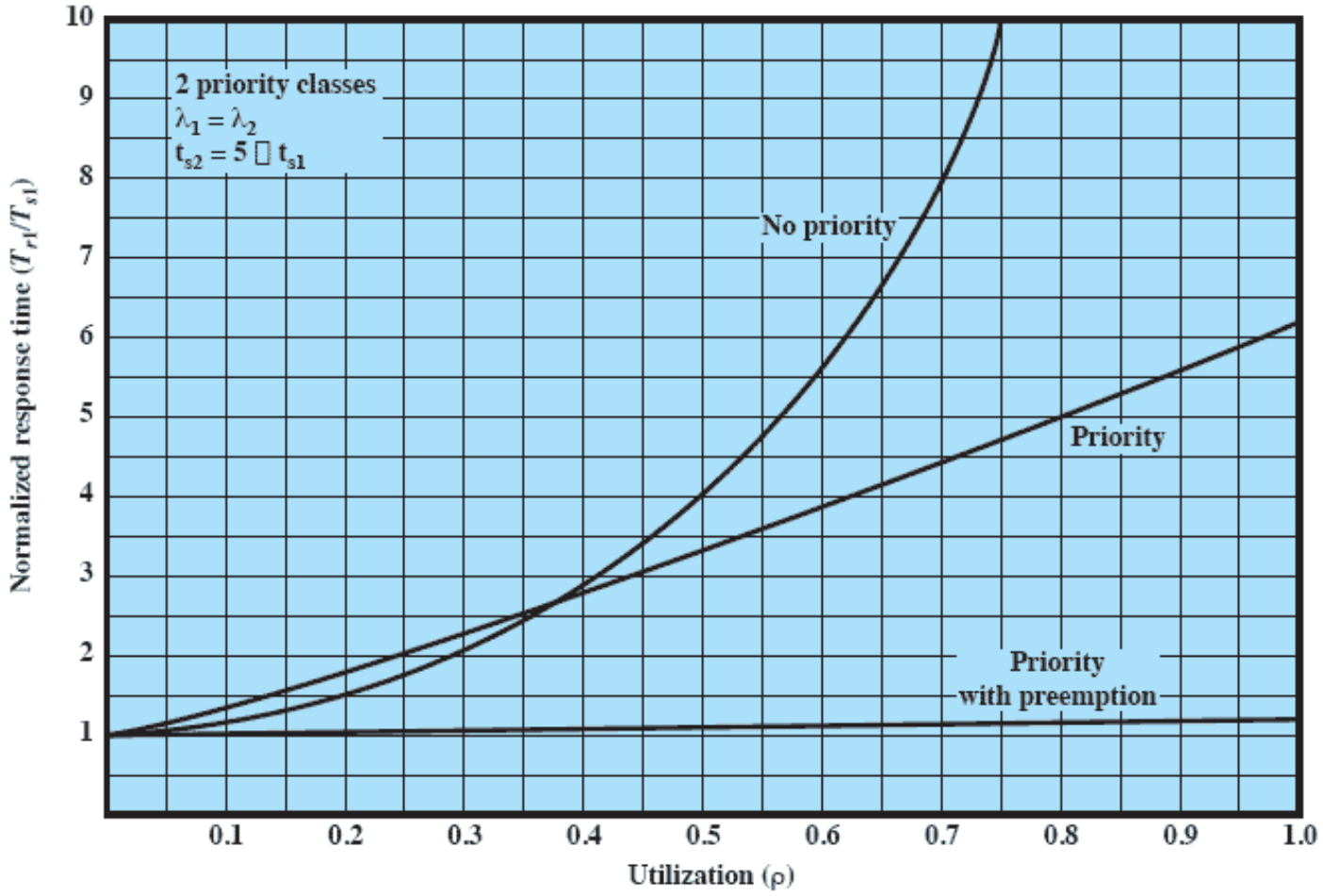


Figure 9.12 Normalized Response Time for Shorter Processes

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Normalized Response Time for Longer Processes

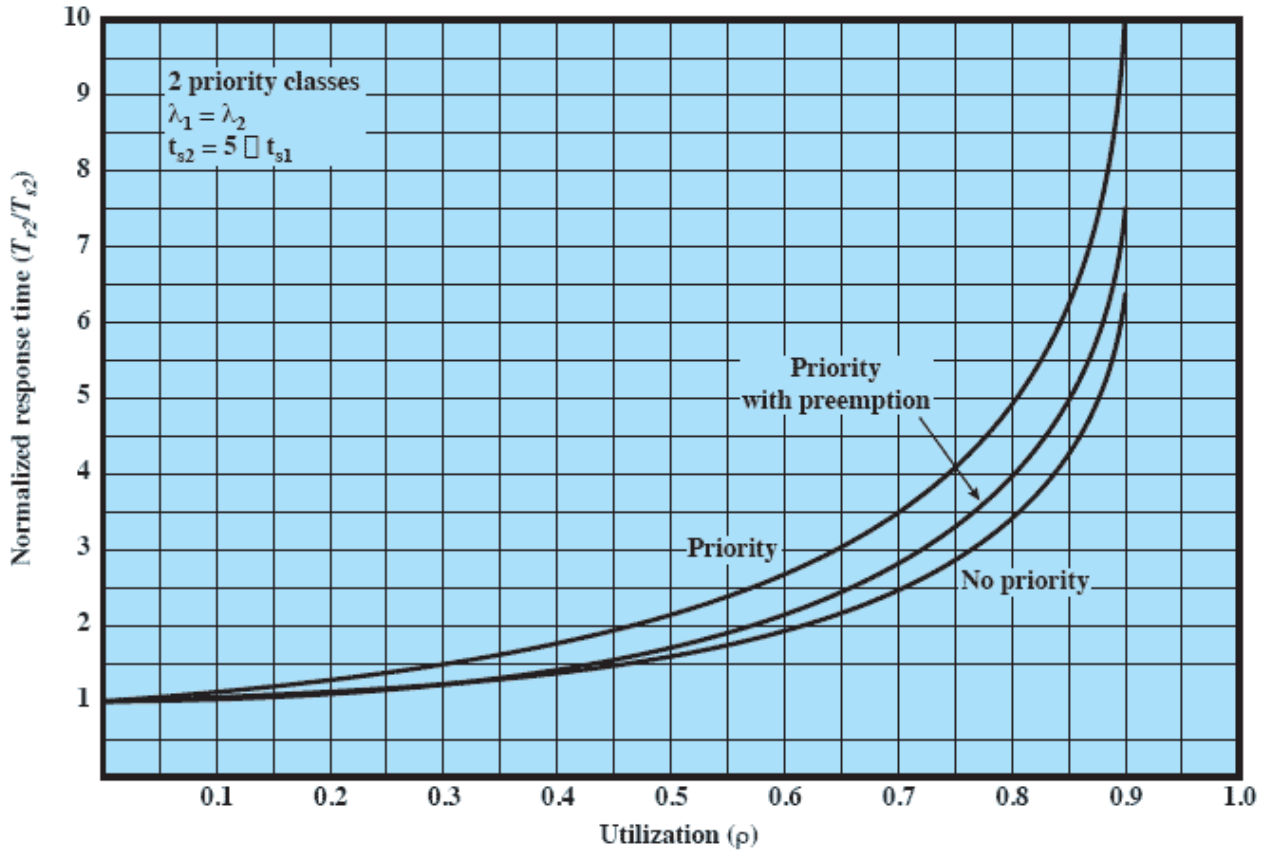


Figure 9.13 Normalized Response Time for Longer Processes

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Simulation Results

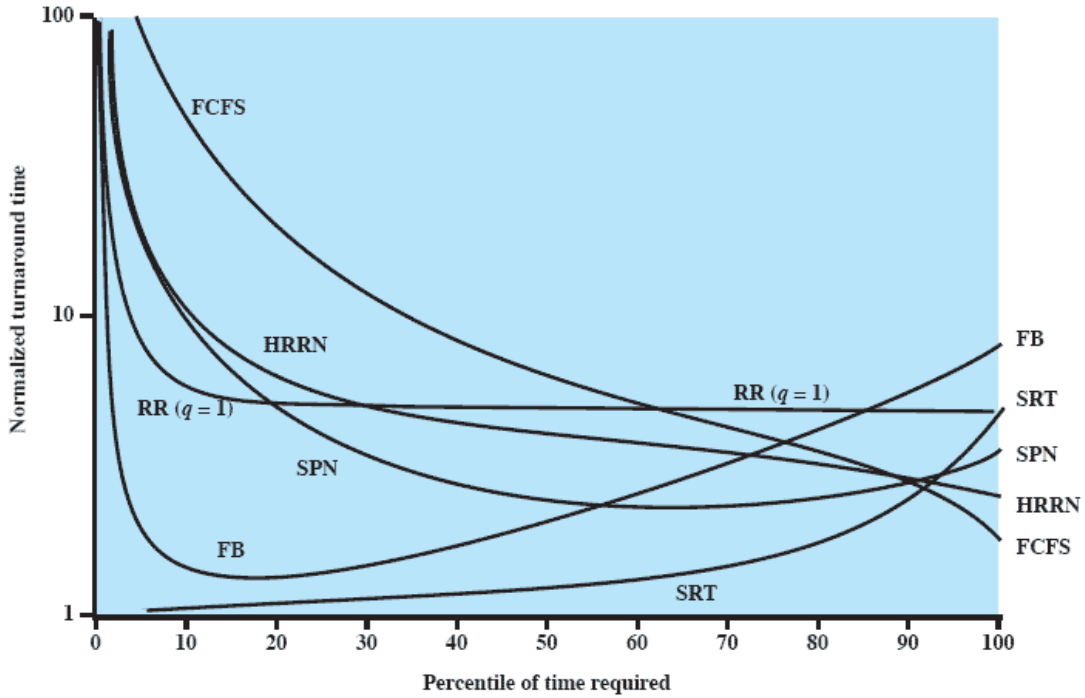


Figure 9.14 Simulation Results for Normalized Turnaround Time

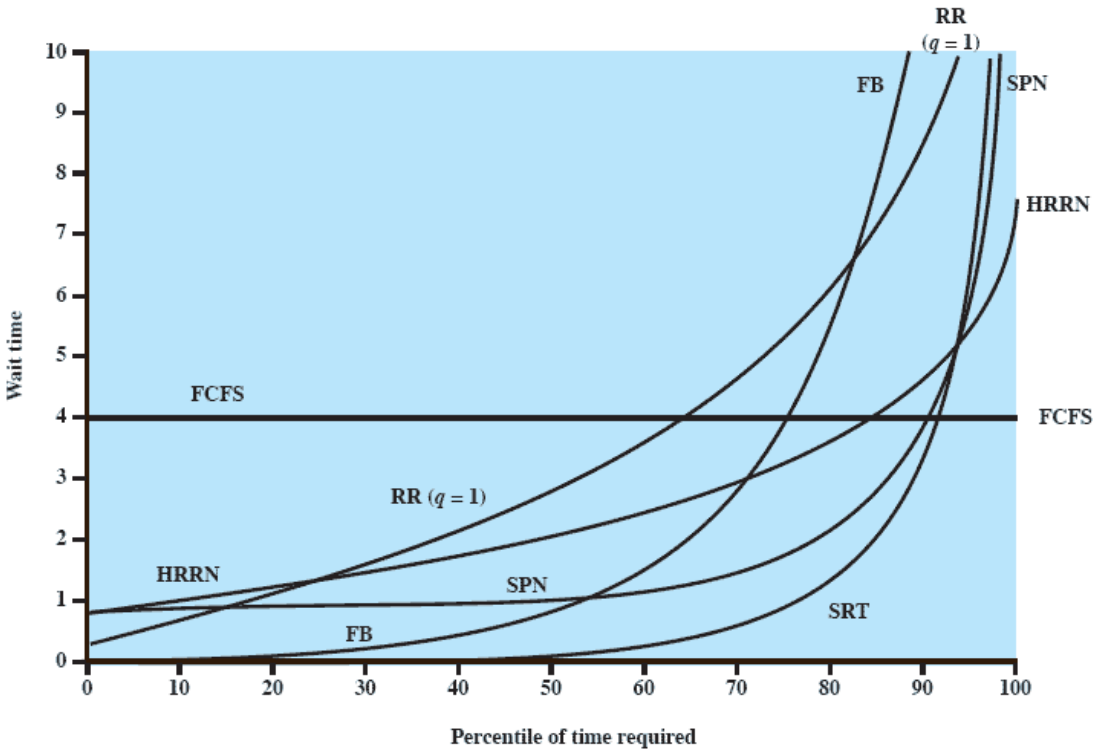


Figure 9.15 Simulation Results for Waiting Time

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Comparison of Scheduling Policies

Process	A	B	C	D	E	
Arrival Time	0	2	4	6	8	
Service Time (T_s)	3	6	4	5	2	Mean
FCFS						
Finish Time	3	9	13	18	20	
Turnaround Time (T_r)	3	7	9	12	12	8.60
T_r/T_s	1.00	1.17	2.25	2.40	6.00	2.56
RR $q = 1$						
Finish Time	4	18	17	20	15	
Turnaround Time (T_r)	4	16	13	14	7	10.80
T_r/T_s	1.33	2.67	3.25	2.80	3.50	2.71
RR $q = 4$						
Finish Time	3	17	11	20	19	
Turnaround Time (T_r)	3	15	7	14	11	10.00
T_r/T_s	1.00	2.5	1.75	2.80	5.50	2.71
SPN						
Finish Time	3	9	15	20	11	
Turnaround Time (T_r)	3	7	11	14	3	7.60
T_r/T_s	1.00	1.17	2.75	2.80	1.50	1.84
SRT						
Finish Time	3	15	8	20	10	
Turnaround Time (T_r)	3	13	4	14	2	7.20
T_r/T_s	1.00	2.17	1.00	2.80	1.00	1.59
HRRN						
Finish Time	3	9	13	20	15	
Turnaround Time (T_r)	3	7	9	14	7	8.00
T_r/T_s	1.00	1.17	2.25	2.80	3.5	2.14
FB $q = 1$						
Finish Time	4	20	16	19	11	
Turnaround Time (T_r)	4	18	12	13	3	10.00
T_r/T_s	1.33	3.00	3.00	2.60	1.5	2.29
FB $q = 2^i$						
Finish Time	4	17	18	20	14	
Turnaround Time (T_r)	4	15	14	14	6	10.60
T_r/T_s	1.33	2.50	3.50	2.80	3.00	2.63

Fair-Share Scheduling

- Bach proposed the Fair Share scheduling strategy in 1986
- Scheduling decisions based on the process sets
- Each user is assigned a share of the processor
- Objective is to monitor usage to give fewer resources to users who have had more than their fair share and more to those who have had less than their fair share

Fair-Share Scheduler

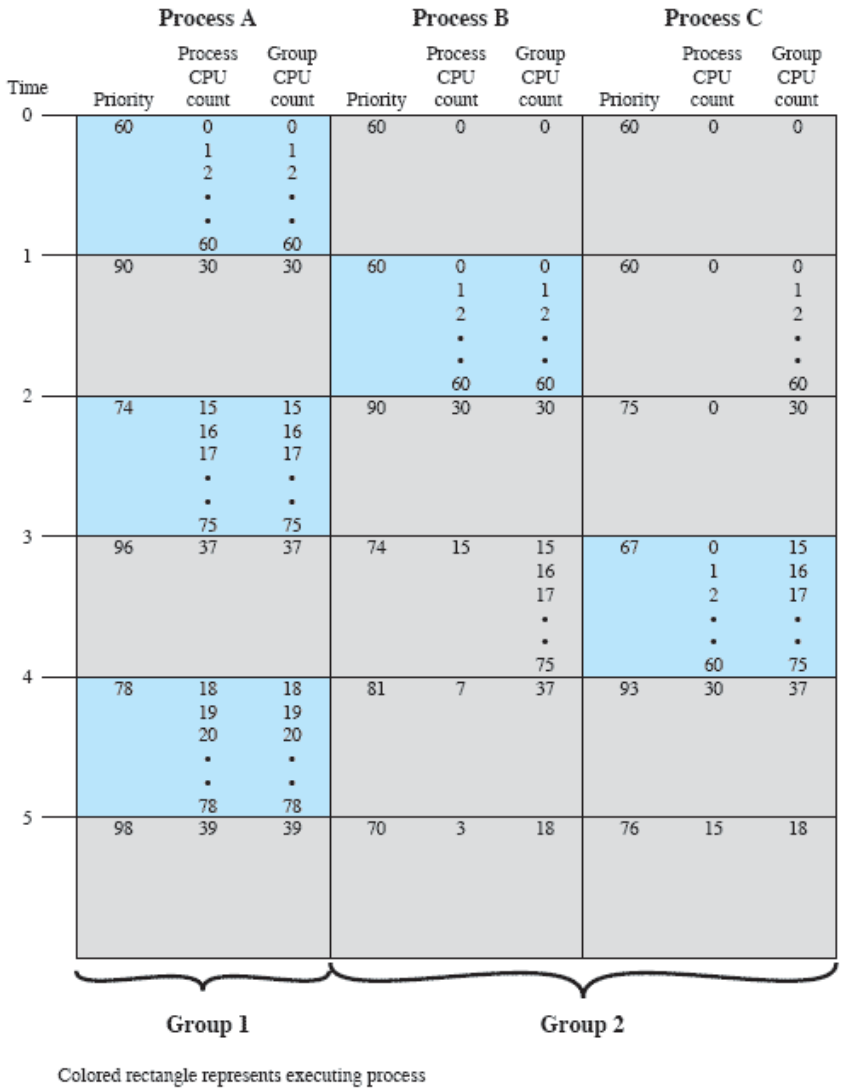


Figure 9.16 Example of Fair Share Scheduler—Three Processes, Two Groups

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Alternative Scheduling Policies

Table 9.3 Characteristics of Various Scheduling Policies

	FCFS	Round robin	SPN	SRT	HRRN	Feedback
Selection function	max[w]	constant	min[s]	min[s − e]	$\max\left(\frac{w+s}{s}\right)$	(see text)
Decision mode	Non-preemptive	Preemptive (at time quantum)	Non-preemptive	Preemptive (at arrival)	Non-preemptive	Preemptive (at time quantum)
Throughput	Not emphasized	May be low if quantum is too small	High	High	High	Not emphasized
Response time	May be high, especially if there is a large variance in process execution times	Provides good response time for short processes	Provides good response time for short processes	Provides good response time	Provides good response time	Not emphasized
Overhead	Minimum	Minimum	Can be high	Can be high	Can be high	Can be high
Effect on processes	Penalizes short processes; penalizes I/O bound processes	Fair treatment	Penalizes long processes	Penalizes long processes	Good balance	May favor I/O bound processes
Starvation	No	No	Possible	Possible	No	Possible

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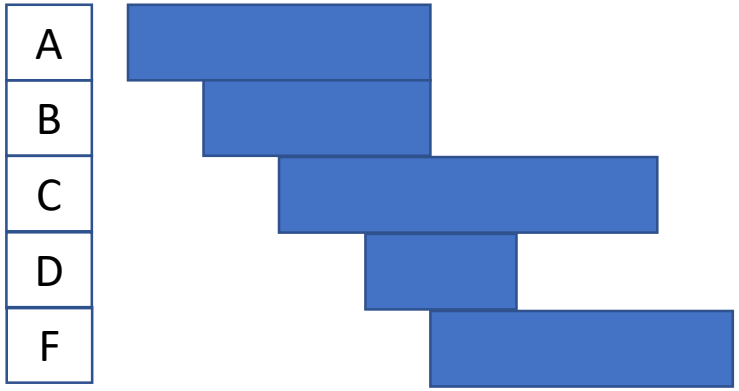
Exercises in class

进程名	A	B	C	D	E
到达时间	0	1	2	3	4
服务时间	4	3	5	2	4

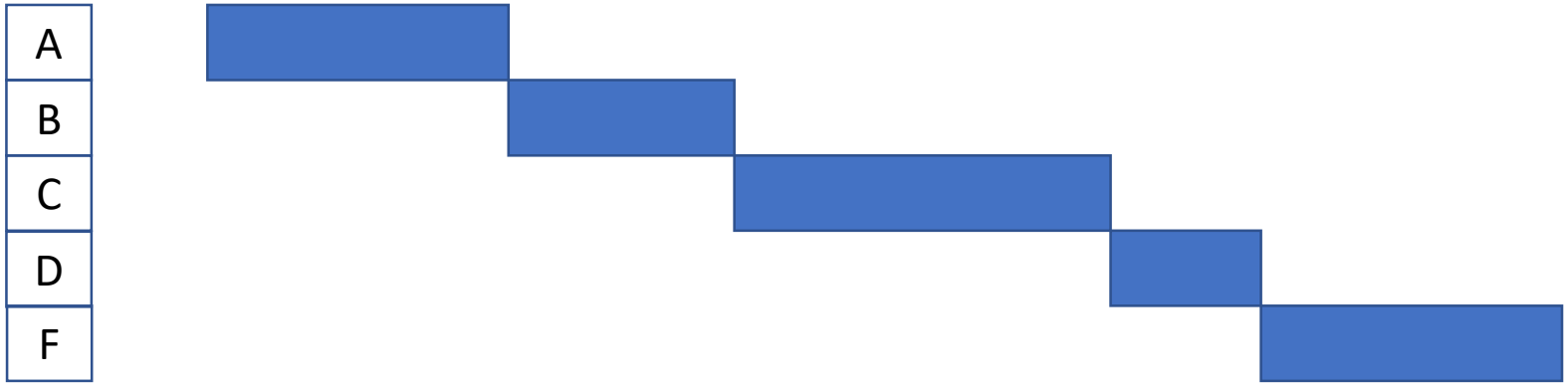
In a uniprocessor,

Use FCFS, RR (q=1?q=3?), SJF, SRT

Request finish time, turnaround time, and draw a Gantt chart.

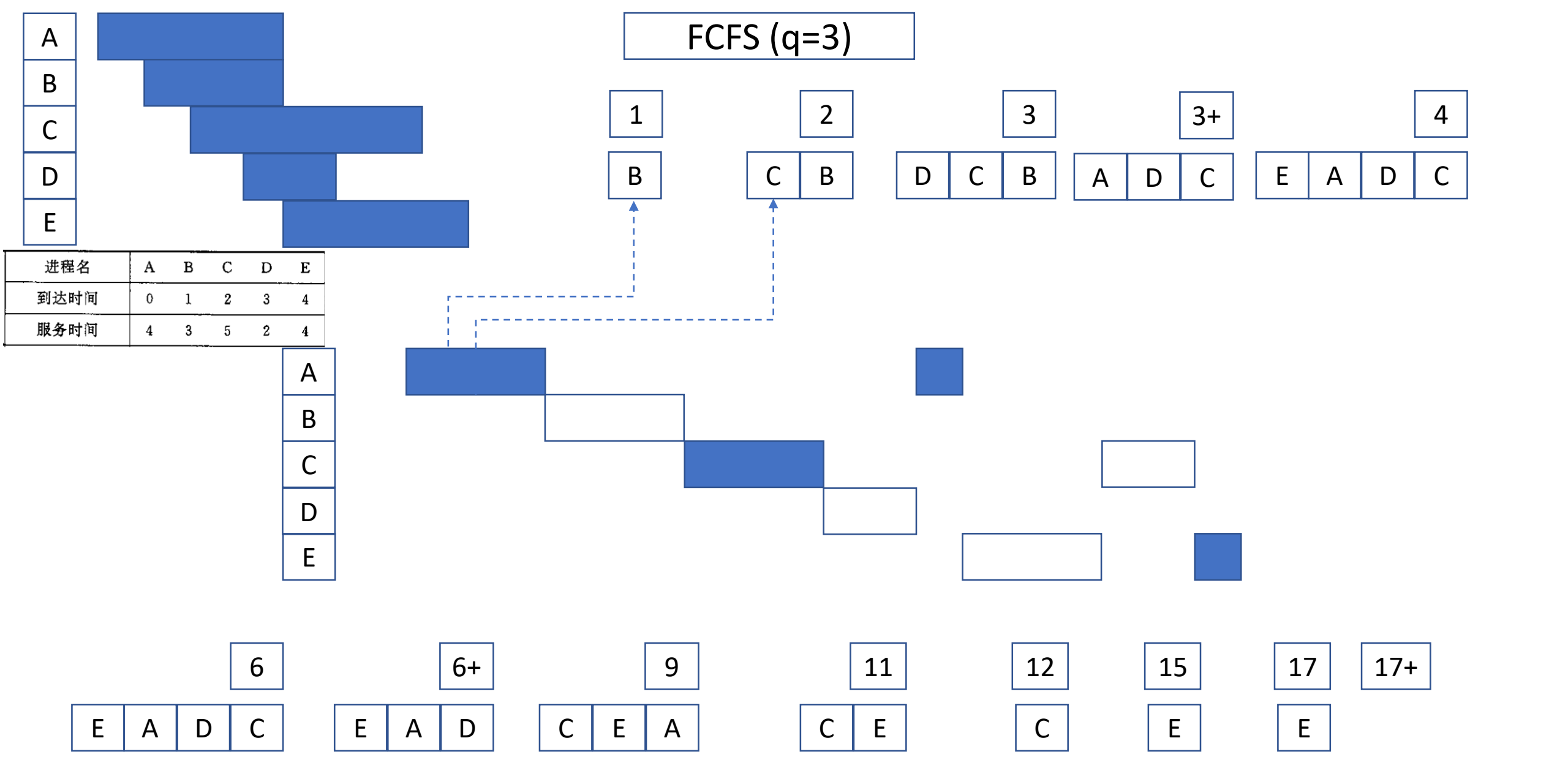


FCFS



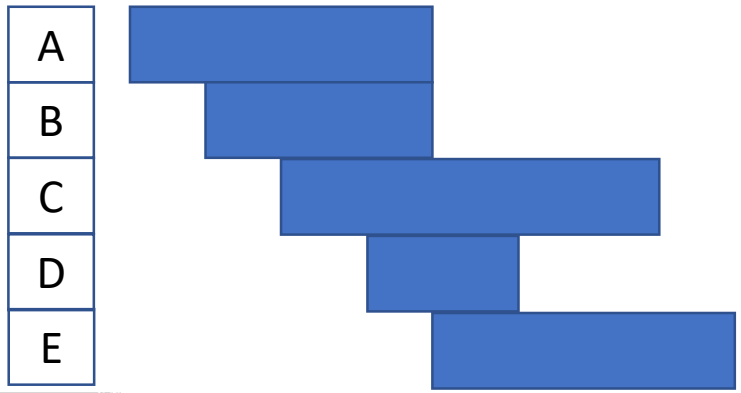
进程名	A	B	C	D	E
到达时间	0	1	2	3	4
服务时间	4	3	5	2	4

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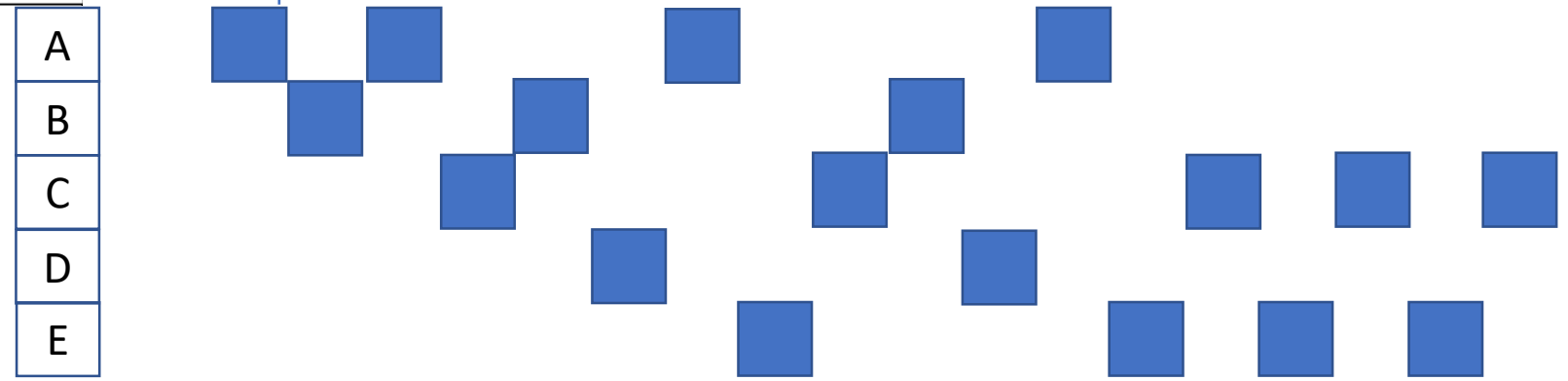
进程名	A	B	C	D	E
到达时间	0	1	2	3	4
服务时间	4	3	5	2	4

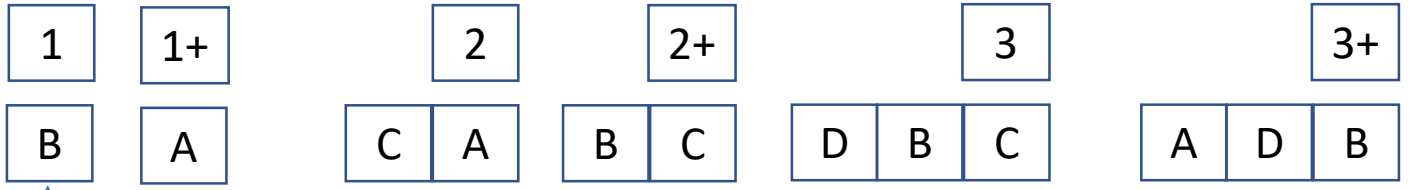
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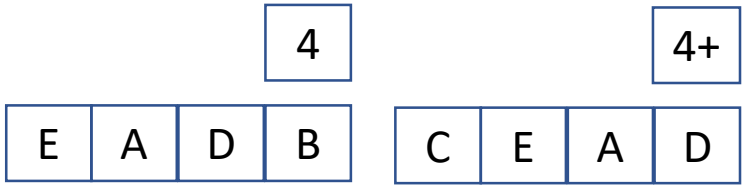


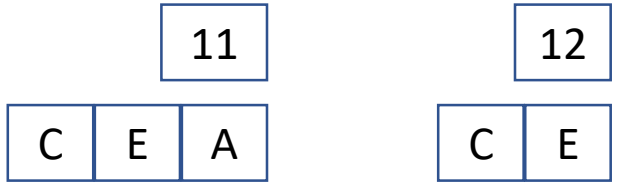
FCFS (q=1)

进程名	A	B	C	D	E
到达时间	0	1	2	3	4
服务时间	4	3	5	2	4

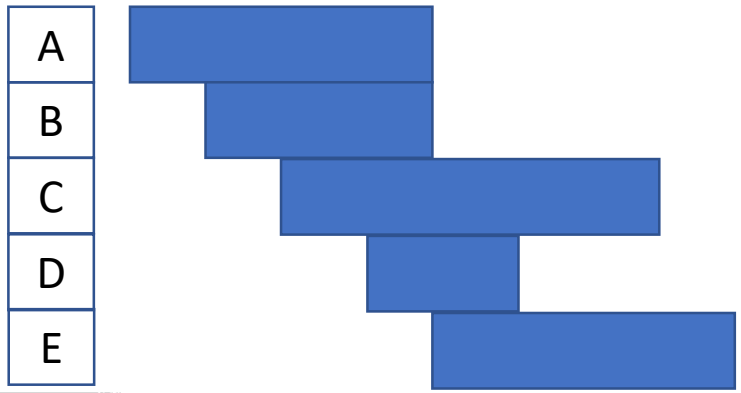








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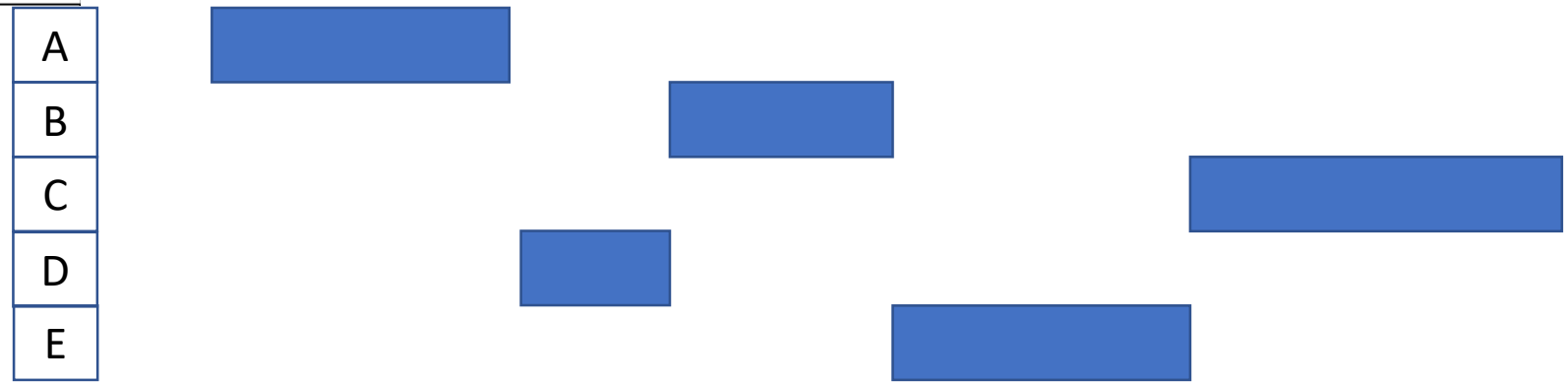


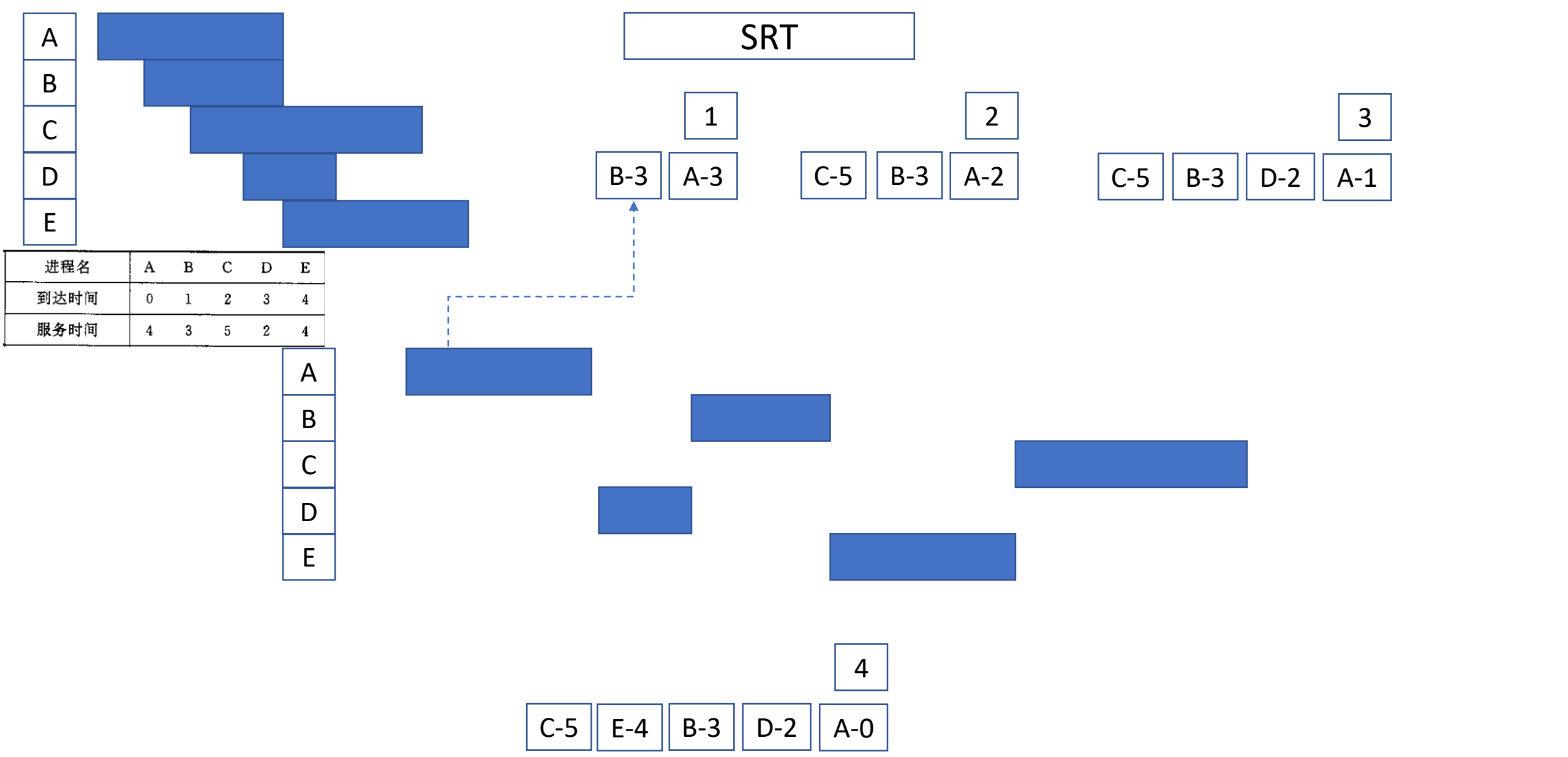
SJT

4

C-5 E-4 B-3 D-2 A-0

进程名	A	B	C	D	E
到达时间	0	1	2	3	4
服务时间	4	3	5	2	4





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What if we can't know the service time in advance?

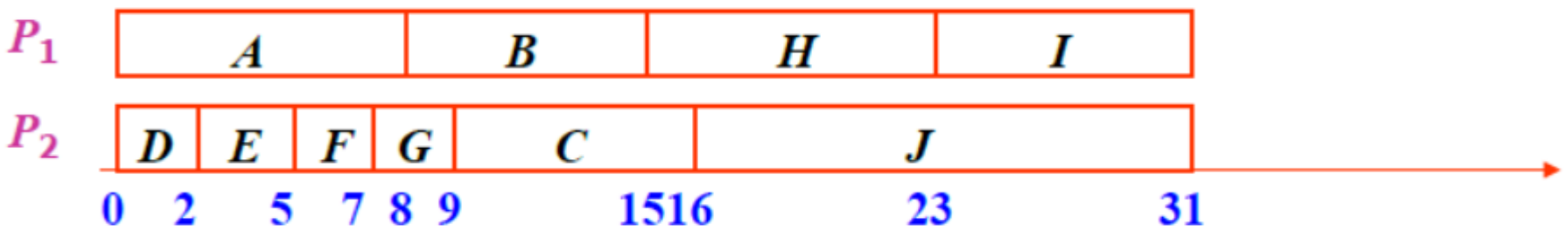
What if in multiple processors with pre-process restrictions?

Example 5　阿克米自行车的装配问题

工　序	紧前工序	加工时间	工　序	紧前工序	加工时间
A	——	8	*F*	*D*	2
B	*A*	7	*G*	*F*	2
C	*A*，*E*	7	*H*	*E*，*G*	8
D	——	2	*I*	*E*，*G*	8
E	*D*	3	*J*	*B*，*C*	15

由两名熟练工人进行装配，要求装完时间最早.

这是一个 $P2|prec|C_{\max}$ 调度问题.



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Summary

- The operating system must make three types of scheduling decisions with respect to the execution of processes:
  - Long-term – determines when new processes are admitted to the system
  - Medium-term – part of the swapping function and determines when a program is brought into main memory so that it may be executed
  - Short-term – determines which ready process will be executed next by the processor
- From a user's point of view, response time is generally the most important characteristic of a system; from a system point of view, throughput or processor utilization is important
- Algorithms:
  - FCFS, Round Robin, SPN, SRT, HRRN, Feedback

Exercise

1. Explain the Preemptive scheduling and Non-Preemptive scheduling?
2. Explain the role of selection function?
3. List and explain the Scheduling Criteria?
4. In a uniprocessor,

   Use FCFS, RR (q=1?q=3?), SJF, SRT, HRRN, FB

   Request finish time, turnaround time, and draw a Gantt chart.

   (skill: queuing over time)

进程名	A	B	C	D	E	F
到达时间	0	2	3	5	7	8
服务时间	8	5	4	2	3	6